# Nonparametric estimation on Riemannian manifolds

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a joint work with Guillermo Henry

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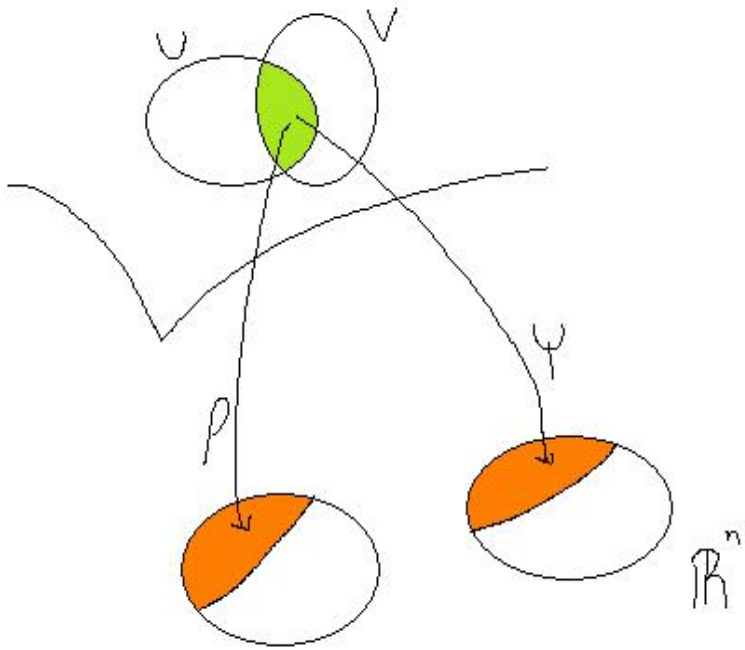
- ( Nadaraya 1964 - Watson 1964)

$$\widehat{r}(x) = \sum_{i=1}^{n} \frac{K(\frac{x - X_i}{h})}{\sum_{j=1}^{n} K(\frac{x - X_j}{h})} Y_i$$

# Preliminaries of Geometry: differential manifold.

A manifold is a topological space which is locally Euclidean.
A differential manifold $M$ is a space with differential structure.




where $\rho$ and $\psi$ are differential functions such that $\psi \circ \rho^{-1}$ is a diffeomorphism.
Example: $\mathbb{R}^n$, cylinder, sphere, surfaces in $\mathbb{R}^n$, lie groups ...

# Preliminaries of Geometry: the tangent space.

For each $p \in M$ we can define the tangent space of $M$ at $p$ that we denote by $M_p$ or $T_pM$ as "the place were the velocities of the curves that pass for $p$ belong.".

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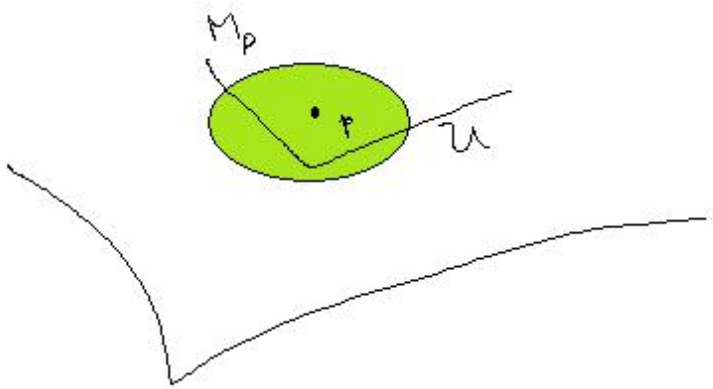


The tangent space is a vectorial space of dimension $n$ and if $(U, \psi)$ is a coordinate system $\{\frac{\partial}{\partial \psi_1}|_p, \ldots, \frac{\partial}{\partial \psi_n}|_p\}$ is a basis of $M_p$.

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Let $p$ and $q$ points in $M$, then $d_g(p, q)$ is the infimum of the lengths of the curves that joint $p$ and $q$.

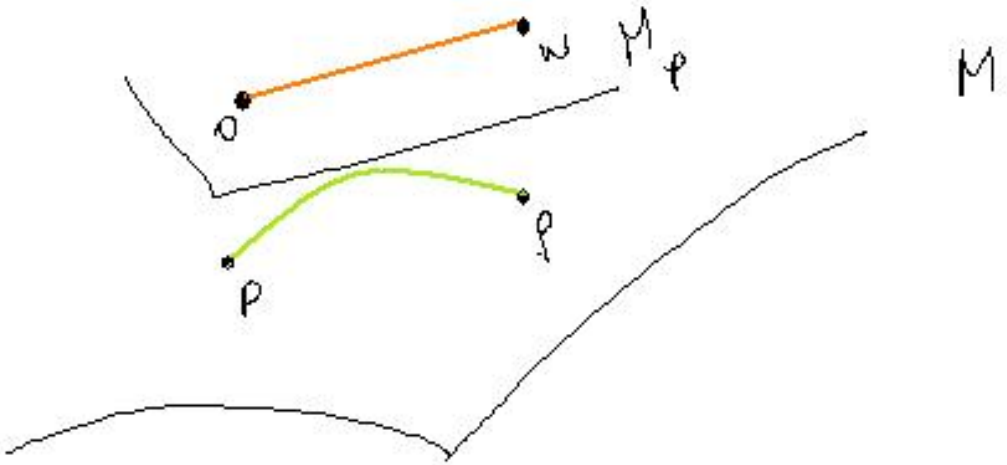
If $(M, g)$ is a completed metric space then there exists a geodesic that makes the distance.
Examples: Cylinder, sphere, the compact manifolds,...

In the following we consider completed Riemannian manifolds.

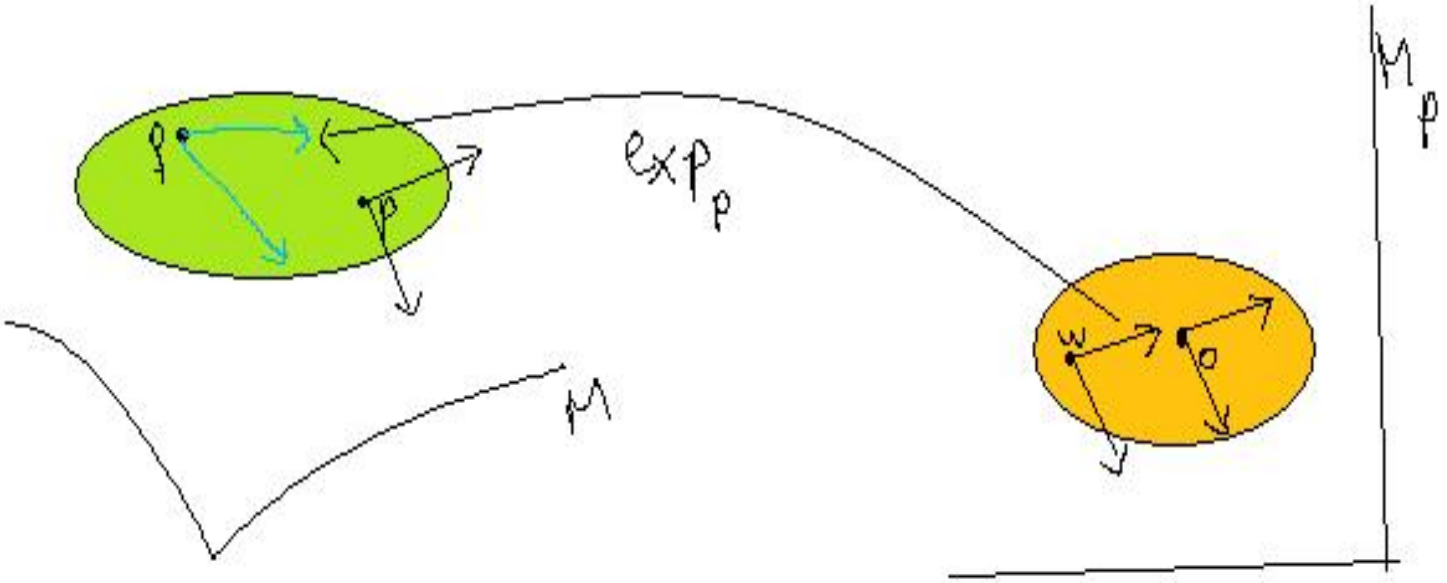
# Preliminaries of Geometry: exponential map.

Let $p \in M$, the exponential map $\exp_p(\cdot) : M_p \to M$ is defined by: for each $w \in M_p$ let the geodesic $c_w(\cdot)$ such that $c_w(0) = p$ and $\dot{c}_w(0) = w/|w|$ the $\exp_p(w) = c_w(|w|)$.

The exponential map transforms lines in $M_p$ into geodesics.



# Preliminaries of Geometry: Volume density function.



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If $w = exp_p^{-1}(q) \in B_s(0_p)$ and we consider the exponential normal chart

$$\theta_p(q) = \left| \det g_q \left( \frac{\partial}{\partial \psi_i}\Big|_q, \frac{\partial}{\partial \psi_j}\Big|_q \right) \right|^{\frac{1}{2}} ,$$

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In general, for any chart

$$\theta_p(q) = \frac{|\det g'\left(\left(\partial/\partial\bar{\psi}_i|_w\right), \left(\partial/\partial\bar{\psi}_j|_w\right)\right)|^{1/2}}{|\det g''\left(\left(\partial/\partial\bar{\psi}_i|_w\right), \left(\partial/\partial\bar{\psi}_j|_w\right)\right)|^{1/2}}$$

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Example in the sphere:

$$\theta_p(q) = \frac{|sen(d_g(p,q))|}{d_g(p,q)} \quad \text{for } q \neq p, -p.$$

Example in the cylinder and $I\!R^d$: $\theta_p(q) = 1$

# Density estimator on Riemannian manifolds

Let $f$ be a density on a Riemannian manifolds $(M, g)$ with dimension $d$. Let $\mathbf{x}_1, \cdots, \mathbf{x}_n$ be a random sample of objects on $M$ with density $f$.

# Density estimator on Riemannian manifolds

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$$f_n(p) = \frac{1}{nh^d} \sum_{i=1}^{n} \frac{1}{\theta_p(\mathbf{x}_i)} K\left(\frac{d_g(p, \mathbf{x}_i)}{h}\right)$$

where $K : I\!R \to I\!R_{>0}$, $\theta_p(q)$ is the volume density function of $(M, g)$, $\lim_{n\to\infty} h = 0$.

# Properties of density estimator

If we consider the case of the sphere, the geodesic distance is,

$$d_d(p,q) = R \ \arccos\left(\frac{< \ p, \ q >}{\|p\| \ \|q\|}\right) = R \ p \sphericalangle q$$

and the volume density function is

$$\theta_p(q) = R \ \frac{|sen(d_g(p,q)/R)|}{d_g(p,q)} \quad \text{if } q \neq p, -p \ \text{ and } \theta_p(p) = 1.$$

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Then

$$f_n(p) = \frac{1}{nh^d} \sum_{i=1}^{n} \frac{p \sphericalangle \mathbf{x}_i}{|sen(p \sphericalangle \mathbf{x}_i)|} K\left(\frac{R \ p \sphericalangle \mathbf{x}_i}{h}\right)$$

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In the case of the cylinder with radius one with the metric induced by the canonical metric of $\mathbb{R}^3$. Let $p, q \in M$, then we can write $p = (\cos(\theta_1), \sin(\theta_1), l_1)$ and $q = (\cos(\theta_2), \sin(\theta_2), l_2)$. The geodesic distance between $p$ and $q$ is,

$$d_d(p, q) = \sqrt{(\theta_1 - \theta_2)^2 + (l_1 - l_2)^2} \quad \text{if} \quad d(\theta_1, \theta_2) < \pi$$

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## Consistency

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with

$$b(p) = \frac{\beta}{2} \int_{\mathcal{V}_1} K(\|\mathbf{u}\|) u_1^2 \, d\mathbf{u} \sum_{i=1}^{d} \frac{\partial f \circ \psi^{-1}}{\partial u_i \partial u_i}|_{u=0}$$

and

$$V(p) = f(p) \int_{\mathcal{V}_1} K^2(\|\mathbf{u}\|) d\mathbf{u}$$

where $\mathbf{u} = (u_1, \ldots, u_d)$ and $(B_h(p), \psi)$ is a exponential chart induced by an orthogonal bases of $T_pM$.

# Regression estimator on Riemannian Manifolds

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$$r_n(p) = \sum_{i=1}^{n} w_{ni}(p)\, y_i$$

where

$$w_{ni}(p) = \frac{\dfrac{1}{\theta_{\mathbf{x}_i}(p)} K\left(\dfrac{d_g(p, \mathbf{x}_i)}{h_n}\right)}{\displaystyle\sum_{j=1}^{n} \dfrac{1}{\theta_{\mathbf{x}_j}(p)} K\left(\dfrac{d_g(p, \mathbf{x}_j)}{h_n}\right)},$$

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- $\sigma(p)$ a robust measure of the conditional scale ($\sigma(p) = \mathrm{mad}_c(p)$).
- $\Psi$ is strictly increasing bounded and continuous functions.

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If $F(y|\mathbf{x} = p)$ is symmetric around $r(p)$ and $\Psi$ is odd, then $r_\Psi(p) \equiv r(p)$.

# Robust estimator of the regression

An estimator for $r_\Psi$

Since

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where $H_n(p)$ is the distance between p and the k-nearest neighbor of $p$ among $\mathbf{x}_1,\ldots,\mathbf{x}_n$ and $k=k_n$ is a sequence such that $k_n \to \infty$

# Robust estimator of the regression

Then, we define the estimators $r_n(p)$ and $\widetilde{r}_n(p)$ as the solution of

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and

$$\sum_{i=1}^{n} \widetilde{w}_{ni}(p)\Psi\left(\frac{y_i - \widetilde{r}_n(p)}{\widetilde{\sigma}_n(p)}\right) = 0.$$

where $\sigma_n(p)$ and $\widetilde{\sigma}_n(p)$ are local robust scale estimators. For instance can be taken as the local median of the absolute deviation from the local median.

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$$r_n^{(\ell+1)}(p) = \frac{\displaystyle\sum_{i=1}^{n} w_{in}(p)\, W\left(\frac{y_i - r_n^{(\ell)}(p)}{\sigma_n(p)}\right) y_i}{\displaystyle\sum_{i=1}^{n} w_{in}(p)\, W\left(\frac{y_i - r_n^{(\ell)}(p)}{\sigma_n(p)}\right)}$$

with $W(t) = \Psi(t)/t$ for $t \neq 0$ and $W(0) = \Psi'(0)$.

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- We iterate until the convergence criterion is achieved.

# Consistency

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$$\sup_{p \in M_0} |\widetilde{r}_n(p) - r(p)| \xrightarrow{a.s.} 0 \text{ as } n \to \infty$$

where $M_0$ is a compact set of $M$

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with

$$b(p) = (\beta^{d+4} f(p)\mu(\mathcal{V}_1))^{\frac{1}{2}} \left\{ \frac{\tau}{6d+12} + \frac{\int_{\mathcal{V}_1} u_1^2 \, d\mathbf{u} \; L(p)}{f(p)\mu(\mathcal{V}_1)} \right\}$$

where $\mathbf{u} = (u_1, \ldots, u_d)$, $\tau$ is the scalar curvature,

$$L(p) = \sum_{i=1}^{d} \left( \frac{\partial^2 f \circ \psi^{-1}}{\partial u_i u_i}\Big|_{u=0} + \frac{\partial f \circ \psi^{-1}}{\partial u_i}\Big|_{u=0} \frac{\partial \theta_p \circ \psi^{-1}}{\partial u_i}\Big|_{u=0} \right)$$

and $(B_h(p), \psi)$ is an exponential chart induced by an orthogonal basis of $T_pM$.

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with

$$\begin{aligned} b_1(p) &= \frac{\beta^{(d+4)/2}}{2f(p)} \frac{\int_{\mathcal{V}_1} K(\|\mathbf{u}\|) u_1^2 \, d\mathbf{u}}{\int_{\mathcal{V}_1} K(\|\mathbf{u}\|) d\mathbf{u}} \; L_1(p) \\ b_2(p) &= b_1(p) \frac{\mu^{1/2}(\mathcal{V}_1)}{f^{1/2}(p)} \end{aligned}$$

$$
\begin{aligned}
V_1(p) &= \frac{\sigma^2(p)}{f(p)} \frac{\int \Psi^2(z) dF_0(z)}{[\int \Psi'(z) dF_0(z)]^2} \frac{\int_{\mathcal{V}_1} K^2(\|\mathbf{u}\|) d\mathbf{u}}{[\int_{\mathcal{V}_1} K(\|\mathbf{u}\|) d\mathbf{u}]^2} \\
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$$

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L_1(p) = \sum_{i=1}^{d} \frac{\partial^2 \phi \circ \psi^{-1}}{\partial u_i u_i} \Big|_{u=0} \text{ with } \phi(q) = f(q)(r(q) - r(p))
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Contaminations:

- ⋆ $C_0 : \varepsilon \sim N(0, 1)$
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Simulation Parameters:

- ⋆ $K(t) = (15/16)(1 - t^2)^2 I(|t| < 1)$.
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- ⋆ $\Psi(u) = (1 - (u/c)^2)^2 I(|u| < c)$.

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The performance of the estimators is measured using an approximation to the $\mathrm{MISE}$ where

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$$M(r_n^i) = \frac{1}{800} \sum_{l=1}^{20} \sum_{j=1}^{40} (r(\mathbf{z}_{lj}) - r_n^i(\mathbf{z}_{lj}))^2,$$

with $\mathbf{z}_{lj}$ a grid of 800 equispaced points in the cylinder.

# Simulation study: Results

| | $h = 0.5$ | | $h = 1$ | |
|---|---|---|---|---|
| | $\mathrm{MISE}(r_{n,\mathrm{c}})$ | $\mathrm{MISE}(r_{n,\mathrm{r}})$ | $\mathrm{MISE}(r_{n,\mathrm{c}})$ | $\mathrm{MISE}(r_{n,\mathrm{r}})$ |
| $C_0$ | 0.4183 | 0.4240 | 0.2015 | 0.2038 |
| $C_1$ | 1832.452 | 2.7432 | 555.7038 | 0.2287 |
| $C_2$ | 459.7590 | 2.1784 | 139.7073 | 0.5069 |

# Simulation study: Bandwidth Selection

Classical cross-validation:

$$CV(h) = \frac{1}{n}\sum_{i=1}^{n}\{y_i - r_{n,c_{-i}}(h)\}^2$$

where $r_{n,c_{-i}}(h)$ is the classical estimator computed with bandwidth $h$ using all the data except $(y_i, \mathbf{x}_i)$. We consider $\widehat{h}_{n,c} = \text{argmin } CV(h)$.

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with $\Psi$ the score function ( ej. Huber). Where $r_{n,\mathrm{r}_{-i}}(h)$ is the robust estimator computed with bandwidth $h$ using all the data except $(y_i, \mathbf{x}_i)$. Then, we choose $\widehat{h}_{n,\mathrm{r}} = \text{argmin } RCV(h)$.

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**Gracias**